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NVIDIA x Red Hat Small Models Hackathon

Agenda

- **Welcome**
- **Weekend Overview**
- **Guidelines, Expectations, & Logistics**
- **Challenge Tracks**
- **Track Technology Spotlights**
  - NVIDIA Nemotron Overview
  - NVIDIA NeMo Gym
  - Red Hat vLLM
- **Bounties & Prizes**
- **Final Reminders**



NVIDIA | Red Hat

Welcome to the Raleigh NVIDIA x Red Hat Small Models Hack



Small models. Big ideas. Real impact.

Build specialized small models, multi-agent systems, and reliable tool-calling solutions—locally or on NVIDIA Brev.

SPECIALIZE

Fine-tune small models to excel at one real-world task.

COLLABORATE

Build multi-agent systems that accomplish more together.

BUILD

Create reliable tool-calling agents that act, adapt, and recover.

SPARK IMPACT

Compete for prizes and showcase what's next in AI.

Hackathon Details

Hackathon Details

The weekend at a glance...

SATURDAY, OCT 17TH

9AM: Doors Open

10AM: Kickoff: Welcome & Hackathon Intro

10:45AM: Brev Coupon Redemption & Team Formation

11AM: Hacking Begins!

12:30PM: Lunch

6PM: Dinner

8PM: Doors Close

SUNDAY, OCT 18TH

9AM: Breakfast

11AM: Code Freeze – Submissions Due!

11:15AM: NVIDIA / Red Hat Developer Roundtables

11:15AM: Judging

12:30PM: Lunch

1:30PM: Awards

Hackathon Details

Guidelines & Expectations

- Don’t CHEAT (existing projects, etc.)
  - All code required to run MUST be written during the hackathon.
  - Open source code is allowed only if it’s been open-sourced for 2+ weeks.
- **Doors close at 8:00 PM and reopen at 9:00 AM tomorrow.**
- Absolutely ZERO drugs or alcohol allowed.
- Treat the space with respect
  - This is an office space 99% of the time.
- Do **NOT** enter into the offices.
- Do **NOT** touch any personal items that may have been left behind at the desks.
- **Clean up after yourself.**

If any conflicts arise or if hackers are found to not be following these guidelines, please find an event organizer to mediate.

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Challenge Tracks



Hackathon Details

Challenge Tracks

All submissions **must** align with one of the three challenge tracks below. Submissions **must** leverage some sort of open data.

Total parameters must be ≤ 33 billion (applies to the total params of all generative language models, including any used for verification, critique, routing, or speculative decoding. Embedding and reranking models are excluded).

Tracks define a **theme of how your AI works, not the scope of your idea** – teams are free to build any solution using any open dataset.

1

FINE-TUNED & SPECIALIZED

A small model that's excellent at one real thing.

Goal: Adapt or fine-tune a small model for a specific job; depth over breadth.

2

MULTI-AGENT SYSTEMS

Many small models beating one big one.

Goal: Wire several small models together so they coordinate to accomplish something no single model could alone.

3

TOOL-CALLING

A single model that uses real tools reliably.

Goal: Deciding when and what to call and falling back gracefully when something breaks.

Check the **Hackathon Notion** for the unique requirements of each track.

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NVIDIA Nemotron Lightning

Seph Mard
Sr. Product Manager, Generative AI Software



NVIDIA Nemotron Models Available Today

Reasoning

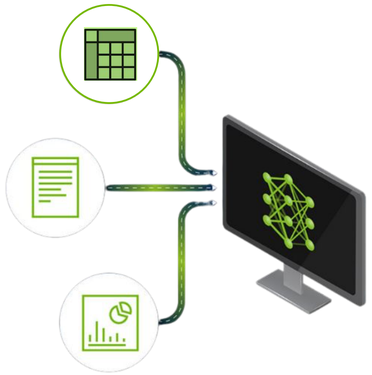


Nemotron 3.5 Lightning
Highest efficiency SLM

Nemotron 3 Super
Highest efficiency LLM

Nemotron 3 Ultra
Frontier-reasoning and orchestration

Vision

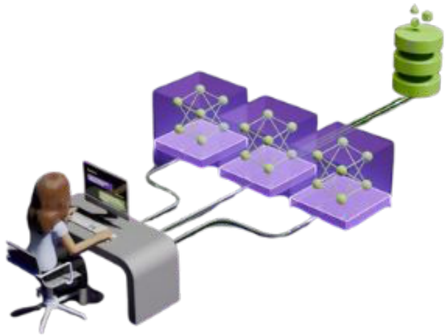


Nemotron Nano 2 VL
Video understanding and Image reasoning

Nemotron Parse 1.1
Document extraction

Nemotron 3 Nano Omni
Enterprise-grade multimodal understanding

Information Retrieval



Nemotron 3 Embed
Leading text embedding model

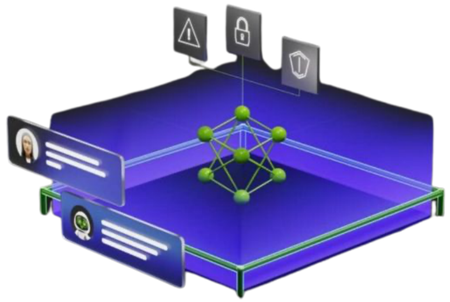
Llama Nemotron Rerank
Multilingual, fine-tuned reranking model

Nemotron Parse
Document understanding and structured extraction

Nemotron Page Elements
Detects text, tables, charts, and figures

Nemotron Table Structure
Reconstructs tables into structured markdown

Content Safety

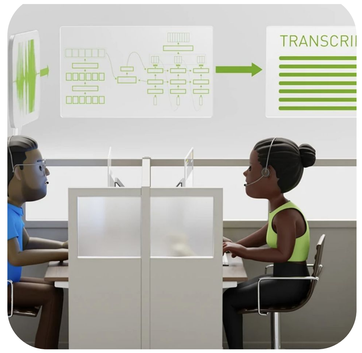


Nemotron 3.5 Content Safety
Multilingual and multimodal guardrails with domain-specific safety policies and reasoning

Llama Nemotron Topic Control
Prevent off-topic / unsafe deviations

Nemotron Jailbreak Detection
Detect adversarial or jailbreak prompts to block unsafe behavior

Speech



Nemotron ASR
Streaming speech recognition in 40 language-locales

Magpie TTS Multilingual
Research-ready, text-to-speech for natural, fine-tunable voices across 12 languages

Nemotron 3 VoiceChat
Full duplex speech-to-speech 12B model built for voice agents
(Early Access)

Nemotron 3 Open Models

Frontier Smart. Faster. More Cost Efficient



Lightning

30B-A3B

AVAILABLE NOW



Super

120B-A12B

AVAILABLE NOW



Ultra

550B-A55B

AVAILABLE NOW

Nemotron 3

Open Datasets

Pre-Training Tokens,
Post-Training Samples, RL Tasks

NeMo Gym

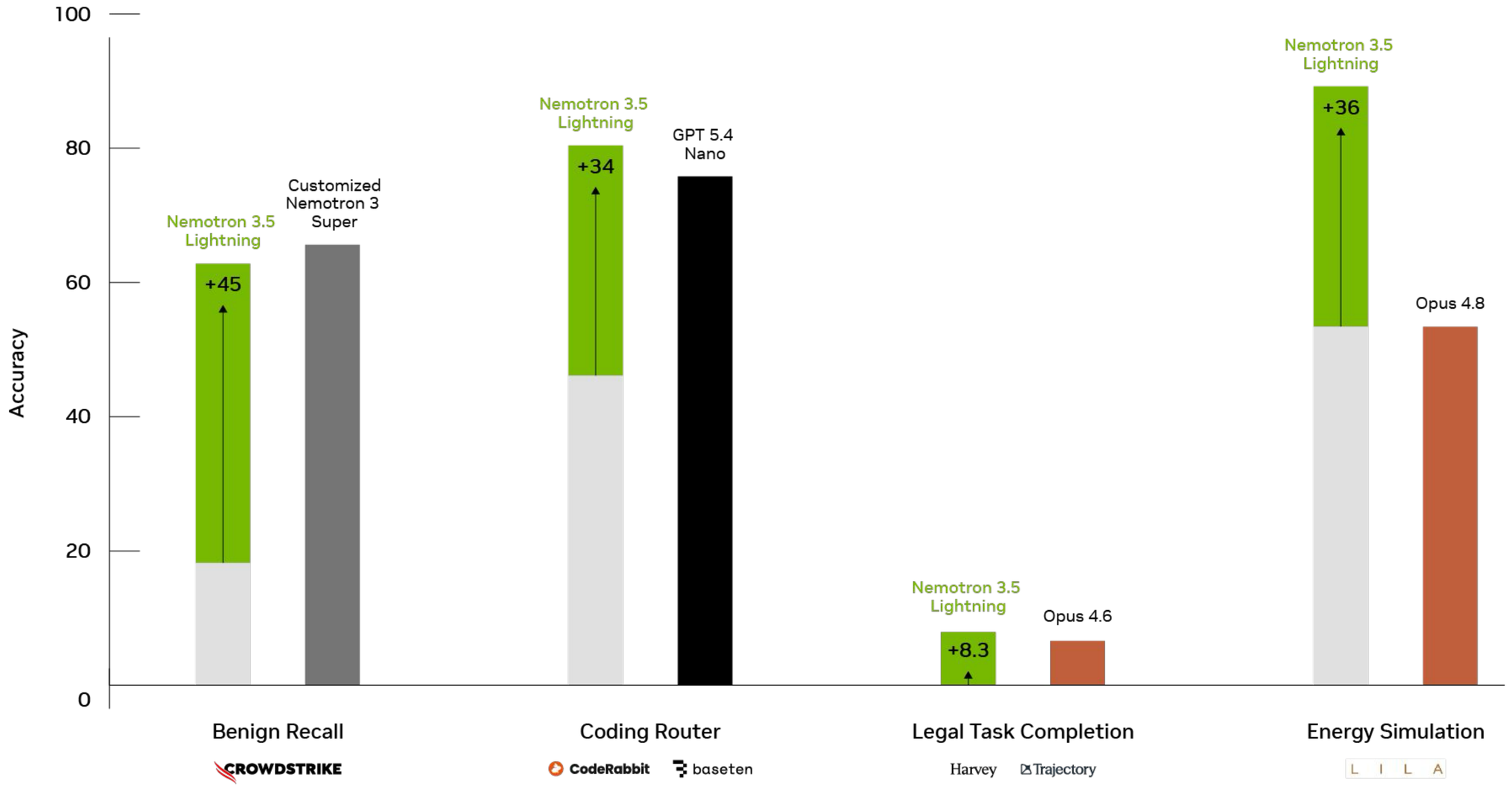
RL Environments & Skills

Open Research

Papers, Samples

NVIDIA.

Nemotron 3.5 Lightning is “Remarkably Easy To Customize”

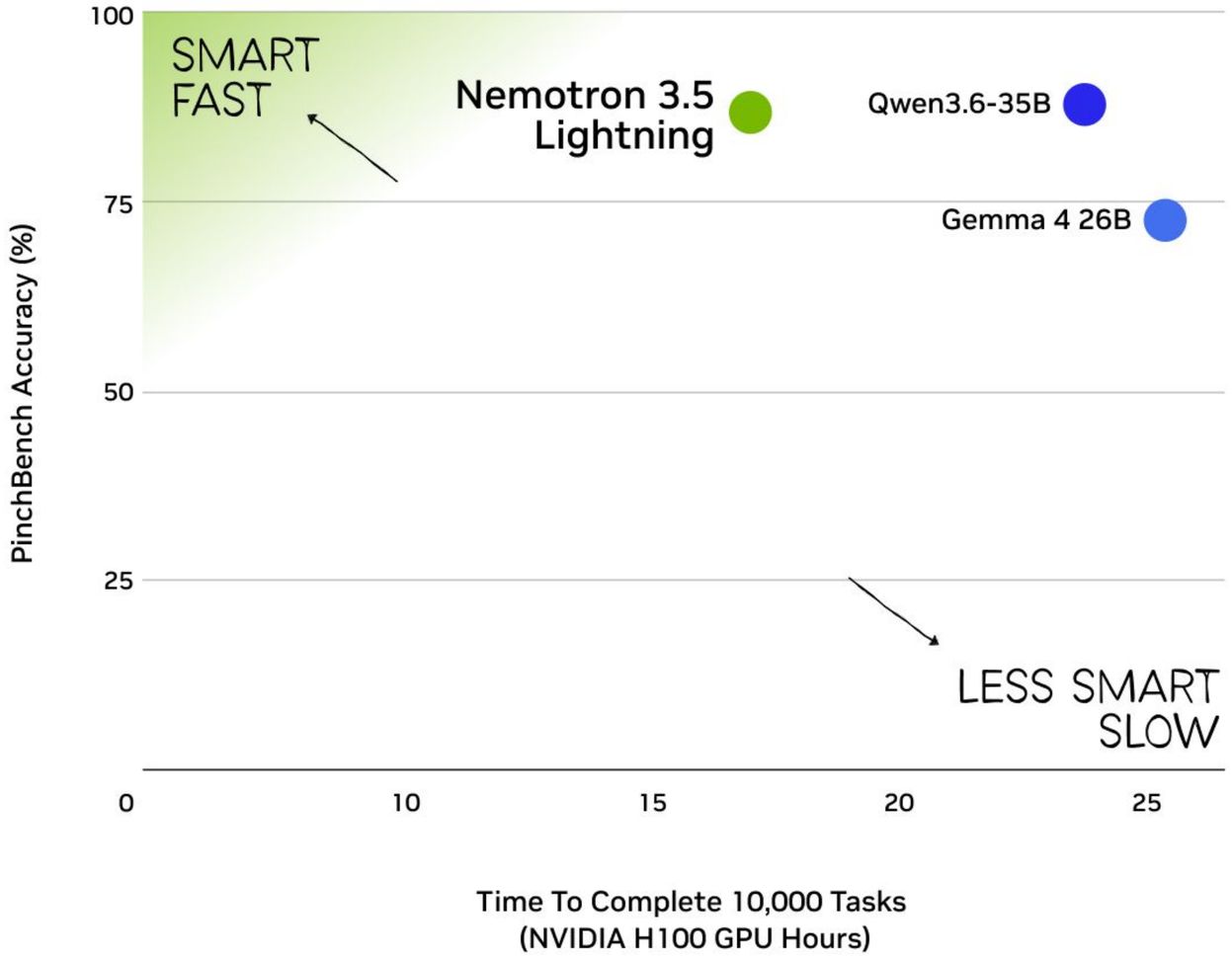


* CodeRabbit

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Nemotron 3.5 Lightning Completes Agentic Tasks 30% Faster, at the Same Frontier Accuracy

Same Intelligence as Qwen3.6 and 30% Faster
Smarter than Gemma 4



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NVIDIA NeMo Gym

Seph Mard
Sr. Product Manager, Generative AI Software



NeMo Gym

Experimentation library for the agentic era



Evaluate and Improve
Shared infrastructure for evaluation, training, and agent improvement



Open and Composable
Integrate with the libraries you already use, and extend your workflows with NeMo Gym's open-source environment hub



Designed for Agentic Era
Modular design to isolate and improve components of agentic systems



Frontier Scale: Production Nemotron Development
2M+ rollouts, 8k+ concurrent rollouts per training step

https://github.com/NVIDIA-NeMo/gym | https://docs.nvidia.com/nemo/gym/about/

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Using Gym: Evaluate → Improve

A systematic loop to measure, diagnose, and improve agentic systems



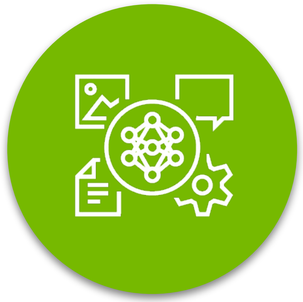
Evaluate

- Benchmark on NeMo Gym environments with verifiable outcomes
- Establish a reproducible baseline across tasks and metrics



Diagnose

- Ablate one component at a time to measure its impact.
- Identify the primary bottleneck across models, prompts, tools, data, or skills



Experiment

- Swap and iterate on models, prompts, harness, tools, data, or skills.
- Re-run the same eval to compare



Improve

- Apply the highest-impact experiment across prompts, tools, skills, harness, or model.
- Post-training, text optimizers, harness improvements.

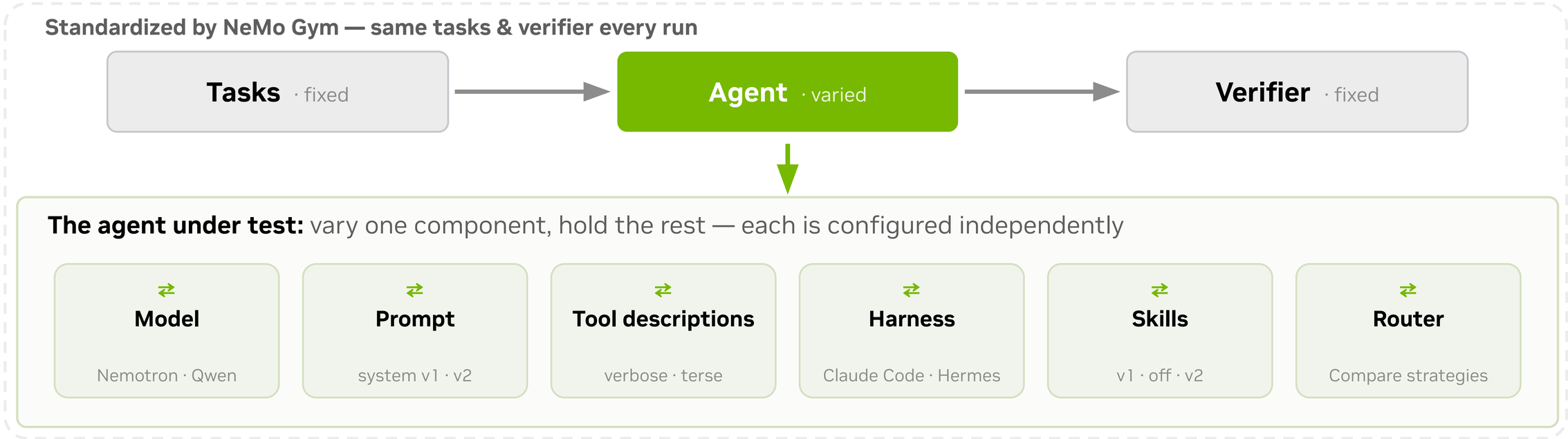
Improvement loop

NeMo Gym is the **environment layer** with **verifiable outcomes** for the agentic era, powering **evaluation**, **improvement**, and **training**.

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Ablate Agent Components

Swap any one component, hold the rest fixed — in NeMo Gym an ablation is a **config change, not a re-build**



WHY EXPERIMENTS ARE EASY IN NEMO GYM

Modular by design

Every component is configured in YAML — swap one by editing config, not rewriting code.

Controlled comparison

Tasks and verifier stay fixed, so any change in the score is attributable to the one part you changed.

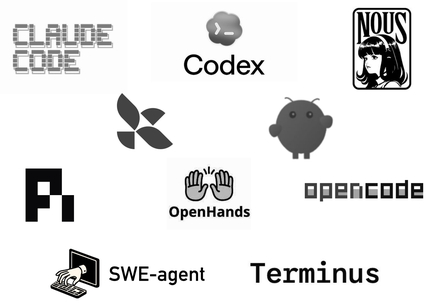
Reproducible

Share the configs to reproduce or compare any experiment exactly — same eval, every run.

Ecosystem Interoperability

Integrate with the tools and libraries you already use

Agent Harness



Environments

aviary harbor OpenEnv

PRIME Intellect reasoning-gym

Training Frameworks

Axolotl miles NeMo/RL

Relax unsloth verl

Inference Software

ANTHROPIC baseten deepinfra

Fireworks AI FriendliAI Hugging Face

NEBIUS OpenAI OpenRouter

SGL together.ai vLLM

Sandbox Providers

APPTAINER AWS Fargate

Daytona docker enroot

OpenSandbox OpenShell

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Red Hat vLLM

Kyle Sayers
Senior ML Engineer



Bounties and Prizes



Hackathon Grand Prizes

Awarded to Each Challenge Track Winner

EACH WINNING TEAM RECEIVES

NVIDIA GeForce RTX 5090

One system per winning team

NVIDIA & Red Hat Swag

High-quality NVIDIA and Red Hat swag for the team

All Things Open Tickets

Free conference passes for
All Things Open Conference 2026



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Best Use of NVIDIA Nemotron Lightning

Bounty

Challenge: Build a project where Nemotron Lightning plays a meaningful role in doing real work. Use it on its own or alongside other models, & show what it enables.

Required: Use the Nemotron Lightning model as a core part of the project.

Demonstrate: Show Nemotron Lightning's contribution to the overall experience, such as quality, speed, efficiency, usefulness, or creativity.

Submit: A short explanation of what Nemotron Lightning is doing, why you chose it for that part of the project, and how you are optimizing its capabilities.

PRIZES: NVIDIA RTX 5070 GPU
NVIDIA Developer Livestream Feature
Social Amplification



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Best Model Improvement With NeMo Gym

Bounty

Challenge: Pick one task a small model or agent fails at today, and make the score better with NeMo Gym.

Required: A NeMo Gym environment with tasks, tools the agent can use, and code that scores each attempt.

Demonstrate: Baseline vs. final scores on the same held-out tasks and eval configuration. Isolate the improvements that made it.

Submit: Your NeMo Gym setup, baseline vs. final results, how to run scoring, and a short explanation of how you made improvements.

PRIZES: NVIDIA RTX 5070 GPU
NVIDIA Developer Livestream Feature
Social Amplification



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Best Use of Red Hat vLLM

Bounty

Challenge: Build a project that is genuinely served on vLLM. vLLM should play a meaningful role in how your application runs, not just be added at the end.

Required: Use vLLM as the serving layer for a core model in your project. Show that your serving setup is functional and central to the build.

Demonstrate: Show how you used vLLM and what you got out of it – such as serving performance, efficiency, throughput, latency, scalability, or reliability.

Submit: Briefly explain how vLLM fits into your architecture, how you configured and optimized it, any measurable results. Depth over breadth.

PRIZES: JBL Speaker (per team member)
Red Hat Office Hours Feature
Social Amplification



NVIDIA Champions Choice Award

Bounty

A select group of NVIDIA Developer Champions will award this bounty to their favorite project!

PRIZE:
$100 NVIDIA Brev Credits per team member!

About NVIDIA Developer Champions Program

A select cohort of community leaders and passionate builders who lead, teach, and mentor, bringing their developer community along with them



Get $25 Brev GPU Credits for Sharing Your Agent's Experience

Build normally while your agent takes notes!

1 Setup

Join the Discord channel: #small-models-hack-raleigh

2 Start

Give your agent the file shared in the Discord: `friction-log.txt`.

3 Build

Build normally. Your agent notes where it gets stuck.

4 Finish & Share

Once done, say "finish friction log" and DM the report to @Robbie.

First 20 submissions will receive a $25 Brev Coupon Code via Discord DM at the end of the hack!

The **report includes** your project goal, agent harness/model, libraries/docs used, and what worked, failed or helped the agent recover. **It does not include raw chat logs, private code, personal information, or secrets.**

Review or edit the report before sending if you'd like.



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Final Reminders



Hackathon Judging Philosophy & Submissions

Project Submissions

PHILOSOPHY

We are judging ***systems engineering***. A winning project isn't just a deck or a simple API wrapper; it's a functioning system that ingests raw data, processes it, and produces a valuable result.

100 Points Total

1. **Technical Execution & Completeness:** Did you actually build a working, complex system?
2. **NVIDIA & Red Hat Ecosystem Utility:** How effectively and meaningfully does the project leverage NVIDIA and Red Hat technologies?
3. **Value & Impact:** Is the solution actually useful and valuable in the real world?
4. **Innovation & Execution:** Did you push the boundaries of what's capable with the technology?

SUBMISSION CRITERIA & TIMELINE

Projects are due by **11AM on Sunday, October 18th.**

Submission Checklist (must include):

- Link to GitHub repository
- Project Description
- 3-5 Minute Video Demo

Submissions will take ~10-15 minutes, so plan accordingly.

Advice: Record the demo video as soon as possible. **Late submissions not accepted.**

Check the **Hackathon Notion** for the demo video criteria

Weekend Resources

Bookmark / Join These Pages

#small-models-hack-raleigh
Discord Channel

Notion w/Resources +
Guidelines + More


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Stay Connected: NVIDIA Community Events

luma.com/nvidia-events

